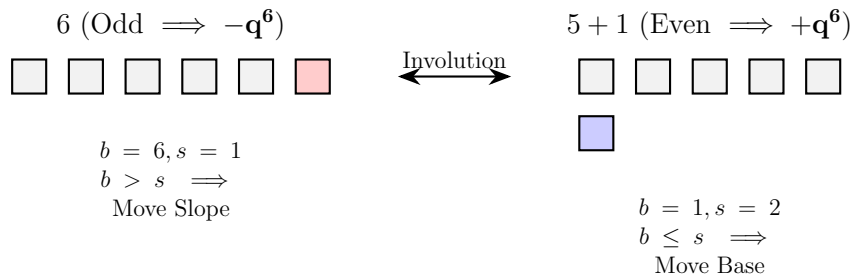


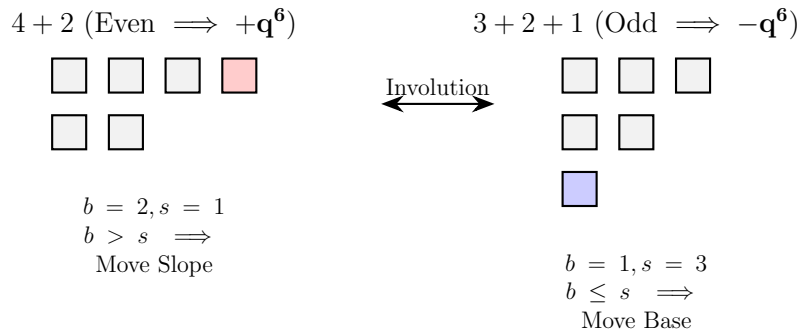
4. The Grand Cancellation: Why the q^6 term is 0

Let's look at all partitions of 6 into distinct parts. There are exactly 4 valid partitions: 6, $5 + 1$, $4 + 2$, and $3 + 2 + 1$. Franklin's involution pairs them up perfectly. The block that moves is highlighted: **Red** when it moves from the Slope, and **Blue** when it moves from the Base.

Pair 1: $6 \longleftrightarrow 5 + 1$



Pair 2: $4 + 2 \longleftrightarrow 3 + 2 + 1$



Summing the components for q^6 :

$$\text{Coefficient} = \underbrace{(-1)}_{\text{from } 6} + \underbrace{(+1)}_{\text{from } 5+1} + \underbrace{(+1)}_{\text{from } 4+2} + \underbrace{(-1)}_{\text{from } 3+2+1} = 0$$

Because 6 is not a pentagonal number, the involution succeeds for every partition, pairing an even parity with an odd parity. The terms annihilate each other, leaving $0q^6$ in the final expansion!